Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C006 **ATP2** 2 genes / 2 species 1 direct pair	**Y1 ATP2** GenAge 29 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 12242224, 18340043	**W1 atp-2** GenAge 25 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11410594, 12471266 17411345, 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C009 **BCY1** 2 genes / 2 species 1 direct pair	**Y1 BCY1** GenAge 42 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 8195187, 20421943	No gene in this candidate network	No gene in this candidate network	**M1 Prkar2b** GenAge 1738 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19536287
Direct pairs: M1–Y1 (n:1)				
C010 **bub-3** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 bub-3** GenAge 900 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18059442	No gene in this candidate network	**M1 Bub3** GenAge 51 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16476774
Direct pairs: W1–M1 (1:1)				
C105 **CDH1** 2 genes / 2 species 1 direct pair	**Y1 CDH1** GenAge 1629 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 fzy-1** GenAge 941 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18059442	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C014 **cdk-7** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 cdk-7** GenAge 107 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	**M1 Cdk7** GenAge 1813 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 22505032
Direct pairs: W1–M1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.